MEKHAZNI Nihad

Telecommunications & Information Technology Engineering Graduate

Aspiring AI/ML Engineer

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SUMMARY

Telecommunications & Information Technology Engineering graduate with academic project experience working with Python, machine learning, and AI technologies. Designed, implemented, and evaluated a Retrieval-Augmented Generation (RAG) system during my graduation project, working with retrieval pipelines, embeddings, reranking, large language models, backend services, databases, and a functional web application prototype. Aspiring AI/ML Engineer interested in building intelligent software applications while continuing to develop my skills through learning and practical projects.

TECHNICAL SKILLS

Programming: Python, Java, SQL, JavaScript, HTML/CSS

AI & Machine Learning: Machine Learning, Scikit-learn, Retrieval-Augmented Generation (RAG), Semantic Search, Hybrid Retrieval, Query Rewriting, Contextual Reranking, Prompt Engineering, LLM Evaluation

Deep Learning & LLMs: Transformers, Embeddings, Reranking, Large Language Models

Frameworks & Databases: LangChain, FastAPI, PostgreSQL, pgvector, Supabase, MySQL, FAISS

Data & Development Tools: Pandas, NumPy, Power BI, Talend, KNIME, Jupyter Notebook, VS Code, Android Studio

PROJECTS

AI-Powered Retrieval-Augmented Generation (RAG) System *Graduation Project | 2026*

- Designed, implemented, and evaluated an end-to-end RAG system for retrieving information from telecommunications technical documentation and generating grounded answers with citations.
- Developed a retrieval pipeline with a query router that determines whether a query requires rewriting, followed by semantic and lexical retrieval, contextual reranking, and LLM-based generation.
- Added an LLM-based verification step to assess whether the retrieved context was sufficient to answer a query before generation, providing a guardrail against unsupported responses.
- Combined semantic retrieval with BM25 using Weighted Reciprocal Rank Fusion to compare and improve retrieval quality.
- Built backend and retrieval components using FastAPI, FAISS, PostgreSQL with pgvector, and Supabase, along with a functional web application prototype featuring authentication, document management, chat history, and citation-based document navigation.

Business Intelligence System *Academic Project*

- Designed a BI pipeline including data warehouse modeling, ETL workflows, dashboards, and predictive analytics.
- Used MySQL, Talend, Power BI, and KNIME for data integration and visualization.

Machine Learning CTR Prediction *AI Competition*

- Applied feature engineering and machine learning models to predict advertisement click-through rates.
- Evaluated and compared different approaches using a real-world dataset.

EXPERIENCE

Technical Intern — Algérie Télécom, Algiers *Jan 2026 – Feb 2026*

- Gained practical exposure to DSL and FTTH access networks, network operations, and service quality monitoring.

Online Intern — Algeria Partners *Jun 2025 – Aug 2025*

- Developed a Python OCR pipeline using image preprocessing and Tesseract OCR for document information extraction.
- Explored data workflows and requirements for a B2B digital marketplace.

Network Intern — Algérie Télécom / ATM Mobilis *2022 – 2025*

- Observed telecommunications infrastructure, network operations, maintenance procedures, and service monitoring.

EDUCATION

State Engineer's Degree in Telecommunications & Information Technology *2021 – 2026* ENSTICP, Algeria

Specialization: Telecommunication Systems and Networks

LANGUAGES

Arabic (Native) · English (Professional Working Proficiency) · French (Professional Working Proficiency)